

# NATHAN VARGHESE

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Boulder, Colorado

## EDUCATION

- MS Aerospace Engineering from University of Colorado, Boulder, GPA - 3.24      Graduated May 2025
- BTech Aerospace Engineering from Maharashtra Institute of Technology, GPA - 3.52      Graduated April 2022

## SKILLS

- **Languages:** Python, C++, MATLAB, Simulink, Julia
- **Software:** Robot Operating Software (ROS1/ROS2), NVIDIA Jetson Linux (Bash, ssh), Gazebo, WeBots, OpenCV, PyTorch, SolidWorks, ANSYS, Fusion 360
- **Dev tools:** Linux systems, Git, Docker, CI/CD, Trello, Testing, TailScale

## WORK EXPERIENCE

Inspector Bots, CO

Robotics Intern

September 2025 - Present

- Leading the development of a Development Kit that integrates the LiDAR, and Pi-camera sensors. Setup using **Docker** and apt **ROS-Humble** image to maintain reproducible environments across multiple devices with the **Jetson Orin Nano** platform. Setup remote ssh using TailScale for a secure remote access to hardware.
- Assist with day-to-day operations of robot **assembly** and **deployment** such as updating firmware, setup configuration, and general troubleshooting for the Roboclaw, RoboteQ, Sabertooth **motor controllers** and analogue radio & video transmitters.

University of Colorado Boulder, CO

Writing Consultant

August 2024 - May 2025

- Provided one-on-one **writing support** to undergraduate through PhD students, including **communication** with both native and **English as a second language (ESL)** speakers, across various disciplines.
- Guided students in strengthening arguments, **organization**, and **clarity** in essays, scholarship proposals, research papers, graduate applications, resumes, and theses.

Teaching Assistant

August 2023 - May 2025

- Held weekly **office hours** and invigilated exams for sophomore and junior level - aerodynamics, statics and structures, and aerospace design lab, aerospace courses.

Maharashtra Institute of Technology (MITADT), India

Research Fellowship

November 2022 - April 2023

- **Designed** and **simulated** the attachment for critical components in SolidWorks using **finite element analysis** in ANSYS, resulted in securing higher research funding from an established research organization.
- Maintained up-to-date technical documentation with **GD&T, assembly & manufacturing drawings, and instruction & maintenance manuals, etc.**
- Managed **communication** between teams across manufacturing, design and with vendors.
- Routinely worked on **slide decks** and **presentations** for updating stakeholders on the status of projects and applications to various research grants.

## PROJECTS

Autonomous AWS Deepracer Competition

September 2024 - December 2024

- Developed the code for a fully autonomous racing car using a motion planning and variable steering control on the Deepracer platform. Conducted extensive testing to fine tune the **Bug algorithm** and the **Proportional-Integral-Derivative (PID)** controller for the racetrack, resulting in the fastest lap and first place in class team challenge.
- Implemented a basic **obstacle detection** and avoidance using the LiDAR data and implemented safety overrides to ensure collision-free navigation. Recorded **rosbag** files for **Rviz**, **SLAM** visualization, setting odometry, and debugging.
- Utilized Python, ROS2, remote ssh, and GitHub for distributing workload and version control. ([Video link](#))